

Submission Summary

Conference Name

2nd International Conference on Advances in Computing, Communication and Networking (ICAC2N2026)

Paper ID

984

Paper Title

Bias Detection and Mitigation in Chatbot Responses Using a Hybrid ML–LLM Framework

Abstract

Bias in the output of large language models (LLMs) presents a major issue for the development of fair and trustworthy conversational systems, with biased content containing stereotypes and assumptions. Current work has mainly addressed bias detection or evaluation, but there is a lack of effective mitigation in a practical and integrated system. In this paper, we introduce a hybrid bias detection and mitigation system that combines a lightweight machine learning classifier with a large language model (LLM) to detect and mitigate text bias. Our system adopts a five-step process comprising input generation, hybrid bias detection, explanation, correction and evaluation. The detection component integrates a transformer-based classifier to detect explicit bias patterns with an LLM-based component to understand context. The correction module then generates unbiased, factual statements while maintaining semantics. An evaluation component using LLM-as-a-judge evaluates the effectiveness of the system in terms of bias removal, meaning preservation, fluency and fairness. The results of the experiments show that the system effectively eliminates bias while achieving similar semantic quality as baseline approaches that fail to remove bias. The system provides an efficient, scalable and adaptable approach to enhancing fairness in conversational AI.

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Submission Files

Bias Detection and Mitigation in Chatbot_LLM.pdf (1.7 Mb, 2026-04-27 12:06:14 GMT+5:30)

Submission Questions Response

1. Name of all Authors

Write Name of all Authors Separated by Comma

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2. Keyword

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6

4. Undertaking

I do hereby undertake that the submitted research paper is my original work and not submitted or published anywhere else.

Agreement accepted
